

Relationships between the Hadley Cell and Subtropical Jet

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Introduction

Recent interest in tropical expansion predicted to occur under a warmer global climate has led to a handful of studies comparing the various metrics used to define the edge of the tropics[◊]. A surprising result from these studies reveal that the location of the subtropical jet (STJ) does not co-vary with the Hadley Cell (HC), as theorized by previous work[‡]. In this research, we perform further analysis of the HC and STJ using data from IPCC's CMIP5 to better understand the relationship between the two atmospheric features in their natural variability and in response to an instantaneous quadrupling of atmospheric CO₂.

References

[◊]Waugh, Darryn W., et al. "Revisiting the relationship among metrics of tropical expansion." *Journal of Climate* 31.18 (2018): 7565-7581.

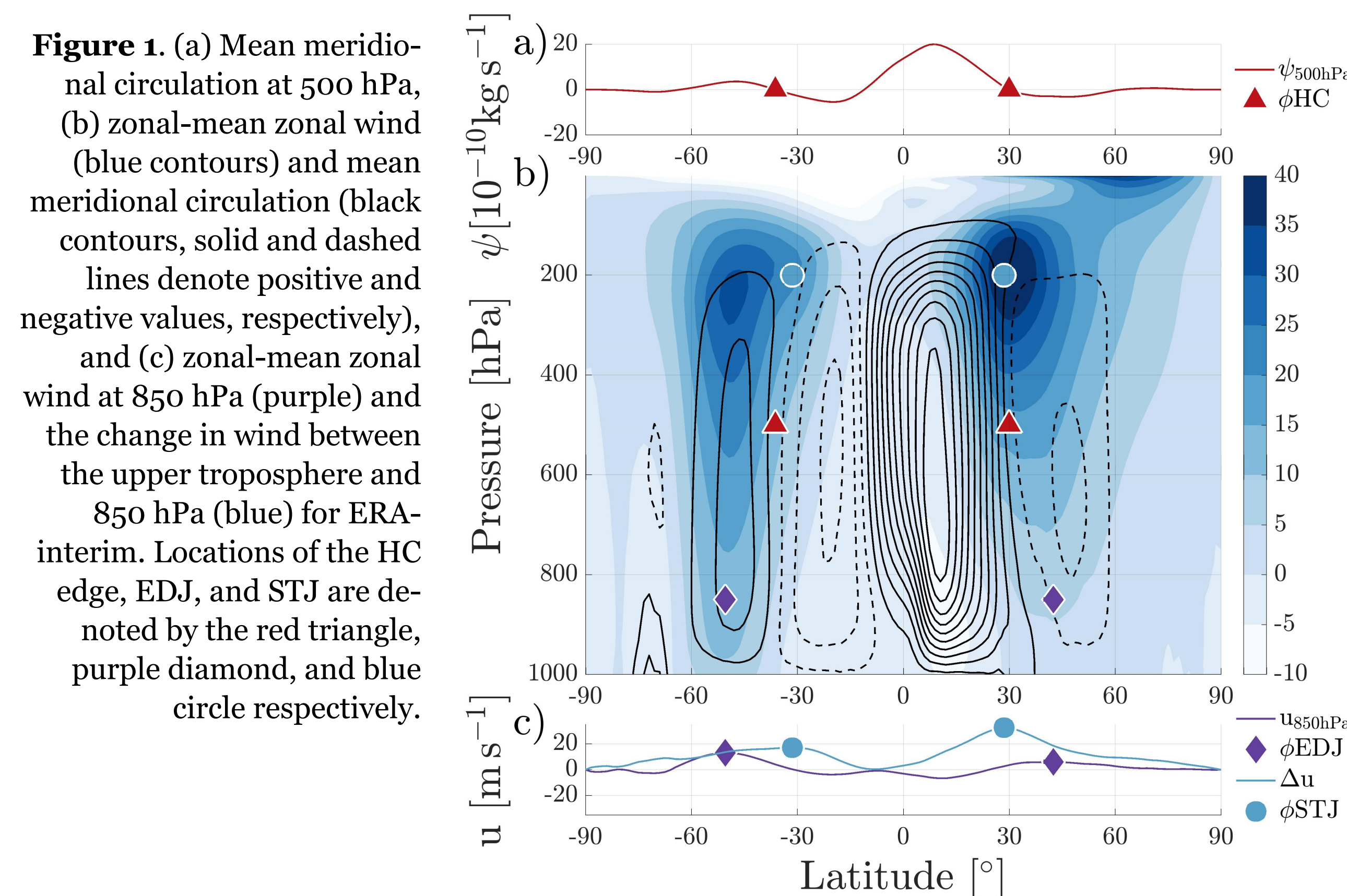
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[‡]Chemke, R., and L. M. Polvani. "Exploiting the abrupt 4xCO₂ scenario to elucidate tropical expansion mechanisms." *Journal of Climate* (2018), in press.

[§]Lu, Jian, Gang Chen, and Dargan MW Frierson. "Response of the zonal mean atmospheric circulation to El Niño versus global warming." *Journal of Climate* 21.22 (2008): 5835-5851.

Conclusion 1: The HC edge and STJ location are not correlated.

Conclusion 2: The STJ strength shows weak correlation with HC, but that relationship differs between natural variability and response to forcing scenarios.



Definition of Metrics

Hadley Cell

$$\phi_{HC} = \phi(\psi_{500 \text{ hPa}} = 0)$$

$$\max_{HC} = \max(\psi_{500 \text{ hPa}})$$

Subtropical Jet

$$\phi_{STJ} = \phi(\max(\Delta u))$$

$$\max_{STJ} = \Delta u(\phi_{STJ})$$

$$\Delta u = u_{100-400 \text{ hPa}} - u_{850 \text{ hPa}}$$

Eddy Driven Jet

$$\phi_{EDJ} = \phi(\max(u_{850 \text{ hPa}}))$$

Analysis

Natural Variability

The HC and EDJ move together but there is little to no correlation between the HC and STJ location.

Poleward,
weaker HC

Weaker
STJ

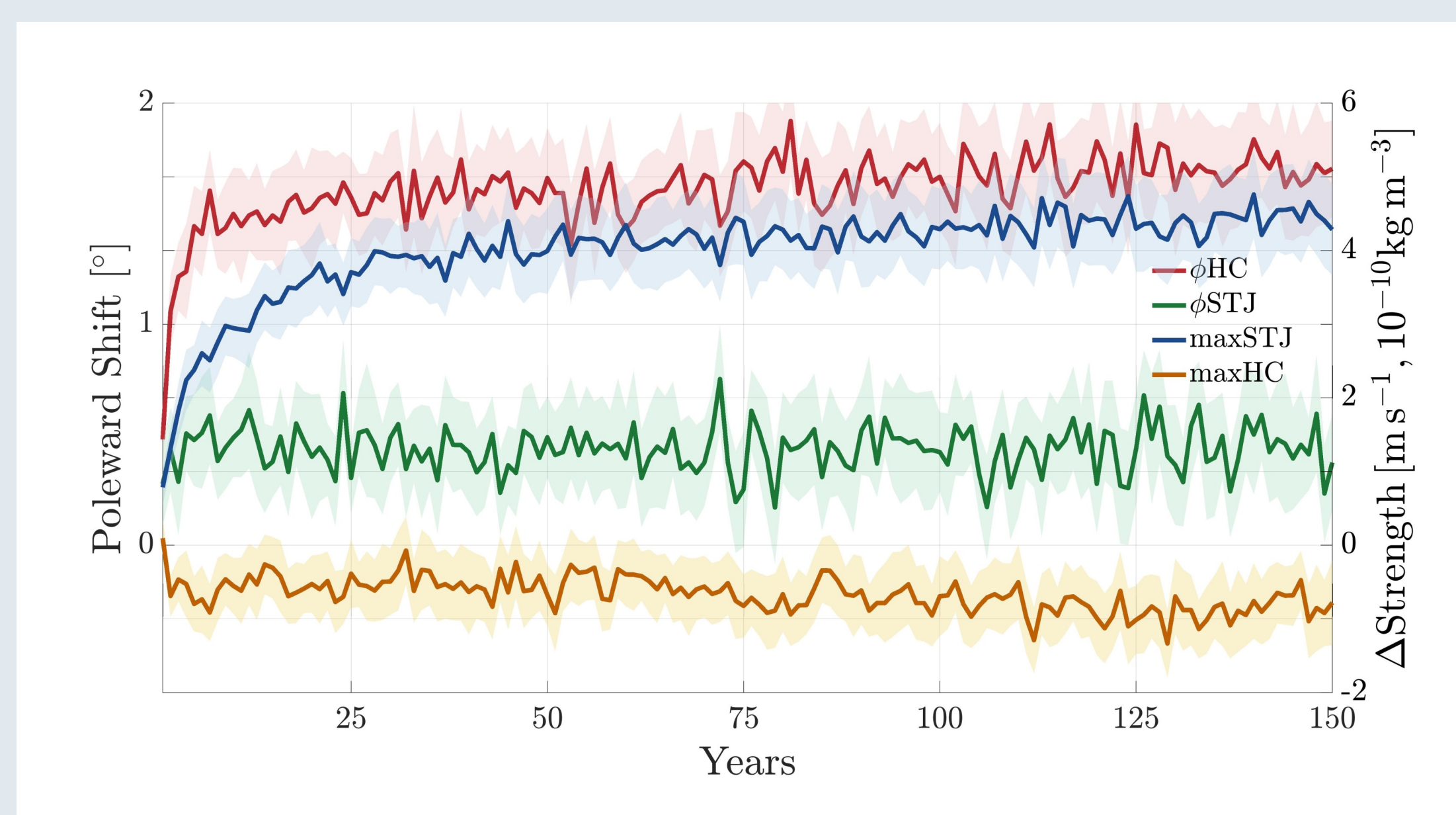


Figure 3. Time series of response to quadrupling atmospheric CO₂ for the HC edge (red), strength (orange) and the STJ location (green) and strength (blue). Response is taken as the difference between abrupt4xCO₂ and the piControl climatology.

Southern Hemisphere

	DJF	MAM	JJA	SON
ϕ_{HC}	0.723	0.462	0.245	0.403
ϕ_{EDJ}	(0.184)	(0.471)	(1.188)	(0.709)
ϕ_{HC}	-0.103	0.102	0.118	-0.029
ϕ_{STJ}	(1.449)	(0.538)	(0.155)	(0.341)
ϕ_{HC}	-0.337	-0.143	-0.25	-0.105
$\max_{STJ}$	(0.338)	(0.22)	(0.409)	(0.235)
$\max_{HC}$	0.057	0.259	0.179	0.222
$\max_{STJ}$	(0.091)	(0.226)	(0.179)	(0.178)
$\max_{HC}$	0.006	-0.107	-0.042	-0.079
ϕ_{STJ}	(0.232)	(0.226)	(0.101)	(0.185)
ϕ_{STJ}	-0.075	-0.208	-0.333	-0.269
$\max_{STJ}$	(0.079)	(0.223)	(0.371)	(0.234)
ϕ_{HC}	-0.005	-0.305	-0.234	-0.355
$\max_{HC}$	(0.169)	(0.186)	(0.273)	(0.194)

Northern Hemisphere

	DJF	MAM	JJA	SON
ϕ_{HC}	0.473	0.484	0.398	0.452
ϕ_{EDJ}	(1.023)	(0.976)	(0.466)	(0.375)
ϕ_{HC}	0.016	0.287	0.2	-0.08
ϕ_{STJ}	(0.144)	(0.27)	(0.292)	(0.321)
ϕ_{HC}	-0.304	-0.521	-0.286	-0.154
$\max_{STJ}$	(0.319)	(0.537)	(0.113)	(0.124)
$\max_{HC}$	0.206	0.395	-0.012	0.118
$\max_{STJ}$	(0.304)	(0.306)	(0.043)	(0.127)
$\max_{HC}$	0.057	-0.078	-0.011	-0.031
ϕ_{STJ}	(0.099)	(0.213)	(0.235)	(0.659)
ϕ_{STJ}	-0.22	-0.391	0.007	-0.034
$\max_{STJ}$	(0.374)	(0.37)	(0.084)	(0.054)
ϕ_{HC}	-0.255	-0.345	0.026	-0.068
$\max_{HC}$	(0.193)	(0.379)	(0.087)	(0.1)

Figure 2. Model-mean correlations in time (bold, colored) and standard deviations (in parentheses) of model spread for various metric relationships. Correlations were calculated with piControl data.

Mechanisms

Time Series of Response

Conclusion 3: The HC poleward shift follows the shift of maximum eddy momentum fluxes* and the STJ strengthening follows the evolution of max meridional temperature gradient, accounting for the difference in evolution.

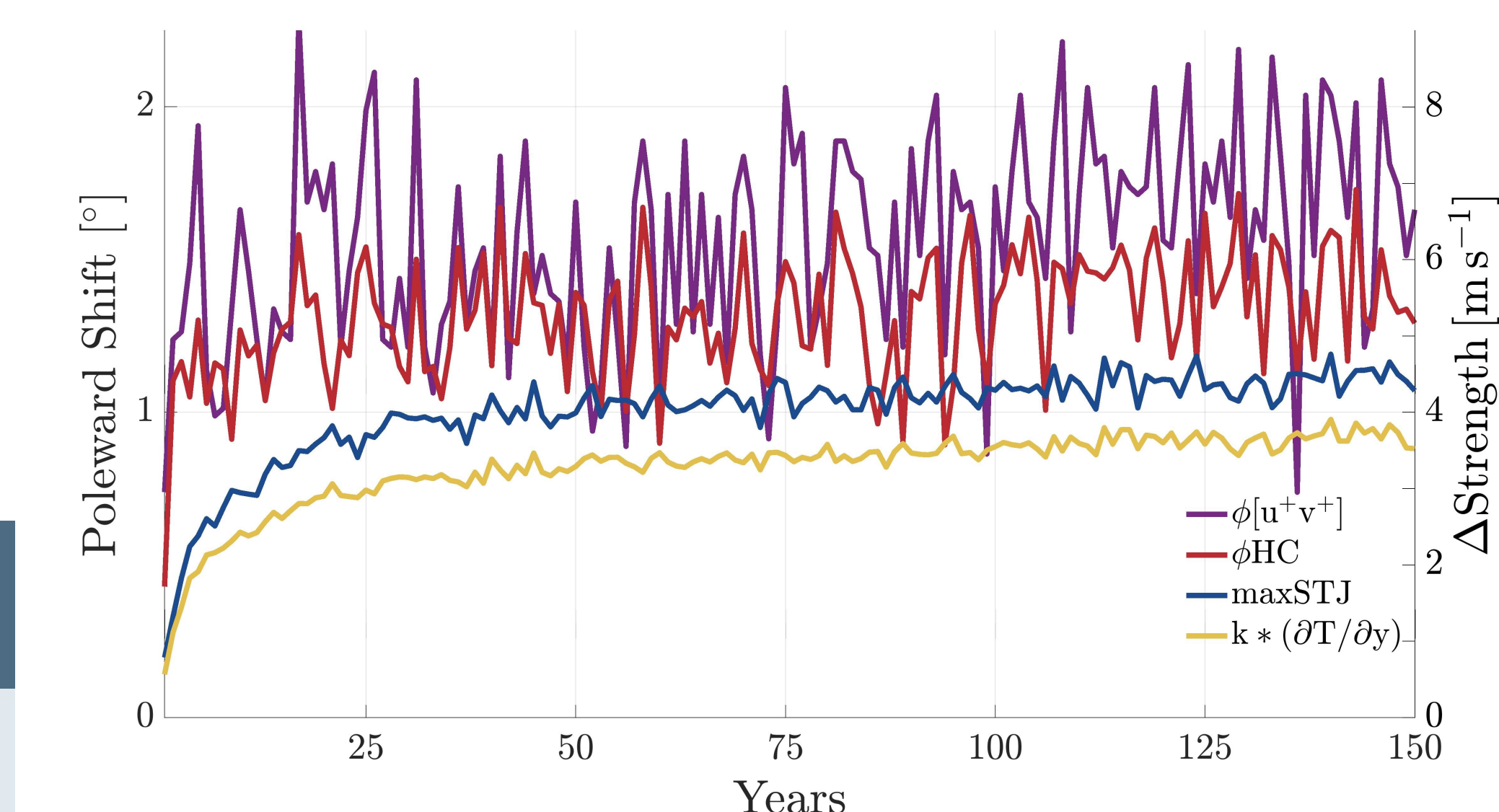


Figure 4. Time series of response to a quadrupling of atmospheric CO₂ for the HC edge (red), location of max eddy momentum flux* (purple), STJ strength (blue) and max meridional temperature gradient (yellow). Response is taken as the difference between abrupt4xCO₂ and piControl climatology. The max meridional temperature gradient is converted to a thermal wind according to:

$$u = -\log \frac{p_r}{p} \left(\frac{R}{f} \right) \frac{\partial T}{\partial y}$$

Natural Variability vs Global Warming

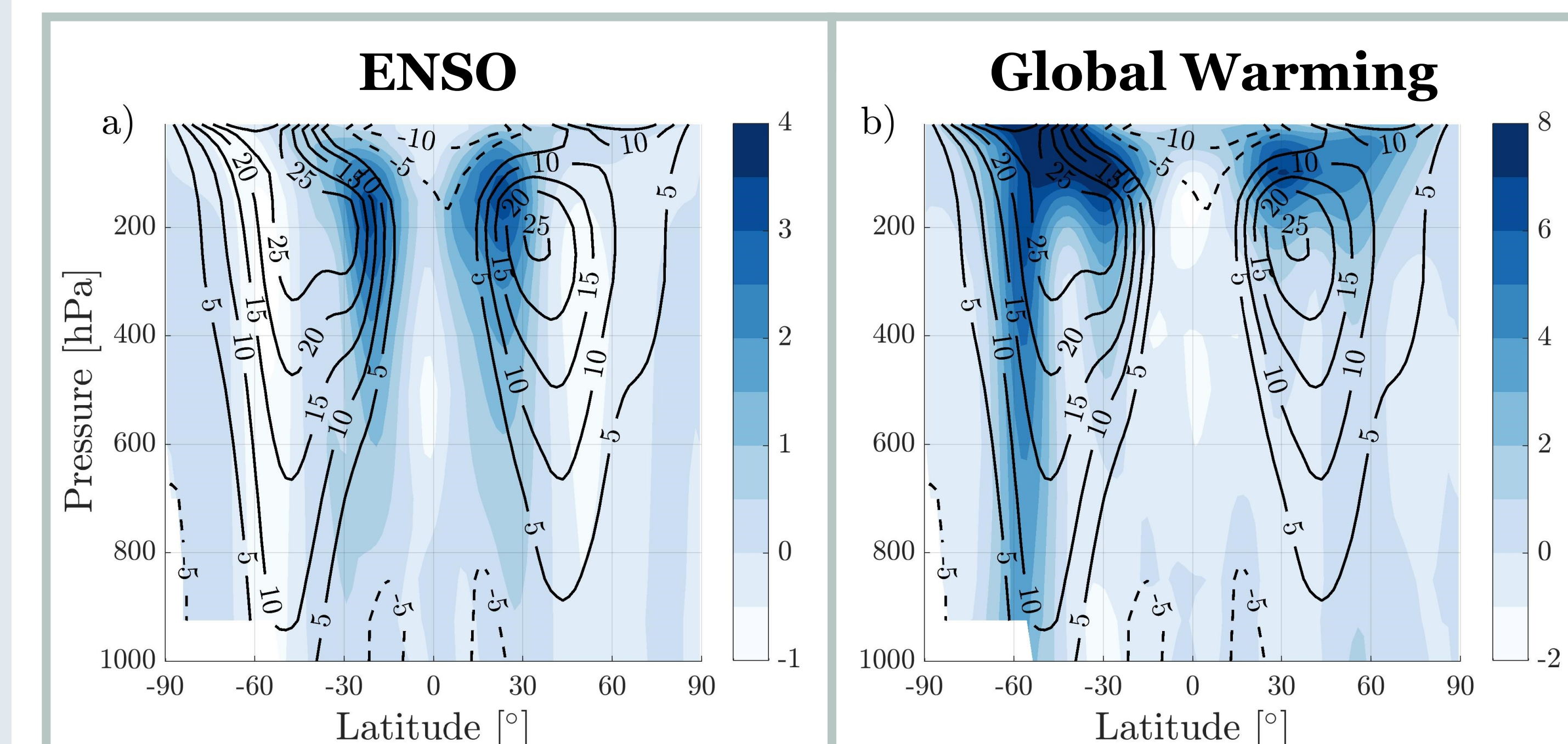


Figure 5. Zonal-mean climatological zonal wind (black contours) and zonal-mean change in zonal wind (blue shading) for (a) ENSO forcing (El Nino minus La Nina) and (b) abrupt quadrupling of atmospheric CO₂ (abrupt4xCO₂ minus piControl climatology) for GFDL-ESM2G. (c.f. Lu et al. 2008[§])

Narrow Warming

HC, EDJ Contracts

Broad Warming

HC, EDJ Expands

STJ strengthens

Response to 4xCO₂

The HC shifts poleward ~1.6° and weakens slightly while the STJ strengthens by about 10% and shifts poleward ~0.4°.

Poleward,
weaker HC

Stronger
STJ

Conclusion 4: The HC and STJ relationship differs between natural variability and global warming due to different responses to narrow vs broader warming[§].